Raspunsuri intrebari MN

curs 13
1. ce inseamna conditie lipschnitz
    tine si pt cond bine pusa
verificare ca are sol unica
metoda euler
metoda taylor
runge-kutta - pe ce se bazeaza
ce inseamna multipas
    +exemple
    cod+ metodav

curs 12
extrapolare richardson
    -ce inseamna
    -de intelegere
legatura romberg si richardson
de unde pleaca romberg(de la metoda trapezului)
    -exemplu +numeric
ce inseamna adaptive
    -de intelegere

-ce puncte consider(cum se aleg punctele)
integrare gaussiana
-ce inseamna
ce inseamna quadratura, grad de valabilitate , care e gradul maxim ce il pot obtine
-numeric (garantat) -
-polinoame legendre
-matlab
-probab mare

curs 11 (derivare)

- legatura polinom lagrange +derivare
{ f’(x) = P_{0, 1} (x) + (x - x0)(x - x1)/2 * f’’(psi(x)) }

formula 3 puncte
derivata de ordin 2 din taylor
derivare de doua ori in punct de mijloc
polinom lagrange
regula trapezelor+simpson

-toate tipurile de subiecte (dedus forma)

-formele compuse

simpson (tot)

curs 10

ce inseamna cele mai mici patrate

-conceptul de aproximare

cea mai buna aproximare -spatii hilbertiene -parte teoretica

- De aratat ca aproximarea e unica (adica g* e unic)

{ ||f - g*|| = min ||f - g||, g apartine subspatiului G).

Fie g1* si g2* cele mai bune aproximatii ale lui f:

<f-g1*,g> = <f-g2*,g> = 0 → g1* = g2* → g* este unic}

- Cum se obtin coeficientii fourier

{c_k = Suma(j = 0, 2m - 1) din y_j * e ^ (j i pi k / m) , pentru k de la 0 la 2m - 1.}

sistem gram
particularizare
-produs scalar
-baza trigonometrica (de facut),
cebasev
aproximari cu fctii rationale
-forma
-de unde scot coeficientii fast fourier transformation
-este aproximare?
-ce legatura are FFT cu aproximarile
- cum se calculeaza acesti coeficienti mai usor?

{In locul evaluarii directe a const **ak** si **bk**, FFT modifica val coef complecsi **ck** in (1/m)*sum(j:0→ 2m-1, ck*exp(i*k*x)) }

cine a facut FFT:
- James Cooley + John Tukey
- care e rezultatul
- cat de bine imbunatateste
{ FFT imbunatateste de la 4 * m^2 operatii, la complexitate O(m logm) }

curs 9

interpolari
-de ce putem face interpolare polinomiala
{ Datorita teoremei Weierstrass }
-lagrange
-toate sub posibile
-ce inseamna interpolari
-diferente divizate
identitatea lui newton
tipuri de diferente cunoscute
ce inseamna df progresive/regresive
-de intelegere
forma newton
ce e u
hermite
-tot
diferenta lagrange/hermite

-cum se obt coef
se mai pune conditie la derivate
de ce nu e buna interpolarea lagrange
spline
-tot
imbinare ce nu e polinom
c2-polinoame de grad 3 de unde se scot conditiile

cum se ajunge la matrici tridiagonale

ce forma are matricea sistemului

curbe bezier

polinoame bernstein

cum se deseneaza

curs 8

q-r

-tot

-ex de facut un pas complet

doar matrice tridiag simetrice

cu deplasare(mereu)

ce asigura deplasarea

ce inseamna deplasare

cum se alege sigma de la deplasare

descompunearea valorilor singulare

-forma

cum se construiesc us

exemplu numeric

Raspunsuri:

curs 9:

Q: Ce inseamna interpolari?

A: Avand un set de n+1 puncte dat, se formeaza un polinom de grad n sau mai mic care sa treaca prin acele puncte.

Q:De ce putem face interpolari?

A: Datorita teoremei lui Weierstrass:
Teorema de aproximare a lui Weierstrass prevede că fiecare funcţie continuă *f (x)* definită pe un interval [*a*, *b*] poate fi aproximată ca limita unui şir uniform convergent de polinoame de interpolare(orice epsilon>0, exista P(x) a.i. |P(x)-f(x)|<epsilon).

3:Polinomul Lagrange:
Aproximeaza functia f cu [sum(k=0->n) f(Xk)]*Lk, unde Lk este multiplicator Lagrange;
L_k=produs(i=0->n,i!=k) (x-xi)/(xk-xi)

4:Dif divizate:pagina 2 slide 12-14
Progresive, regresive pagina 4
Progresive(Inainte) -> $\Delta f(x_i) = f(x_i + h) - f(x_i)$
Regresive(Inapoi) -> $\nabla f(x_i) = f(x_i) - f(x_i - h)$
Centrate -> $\delta f(x_i) = f(x_i + h/2) - f(x_i - h/2)$
http://ro.wikipedia.org/wiki/Diferen%C8%9Be_divizate

5:De ce nu e buna Lagrange?
R:Pentru ca interpolarea cu polinoame Lagrange produce oscilatii mari in aproprierea capatului de interpolare

6:Diferente Lagrange/ Hermite
La Lagrange ai conditia ca polinomul de interpolare sa treaca prin cele n+1 puncte; la Hermite apare conditia suplimentara ca in puncte sa coincida cu derivata functiei in acele puncte; deci trebuie cunoscut f'(xi);

SPLINE

-curba parametrica definita printr-un set de puncte de control
S;[a,b]->R, definta local pe mai multe intervale Pi:[xi,xi+1]->R a=x0..xn;...

clasificare:
Clasa C1:liniare
conditi de interpolare:sa treaca prin setul de puncte dat ca parametru
Si(xi)=f(xi) (i=0:n-1);
Sn-1(xn)=f(xn)
conditii de racordare:P(i-1)(xi-1)=Pi(xi)

pt n pct 2n conditii

C1:cubice
cond de interpolare:la fel ca sus
conditii de racordare: la fel
continuitatea derivatei: in punctele interne (derivate de ordin 1)
2n+2 conditii interpolare
2n-2 pt racordare

C2:cubice
derivata de ordin 2 continua in punctele interne;
conditii de racordare:
n+1 conditii de interpolare
3(n-1) conditii de racordare-continuitatea in pct+in derivatele 1 si 2
=>4n -2 conditii;
pana la 4n mai sunt condtii de splne natural/ tensionat
NATURAL:derivatele de ordin 2 din capete =0
TENSIONAT:derivate de ordin 1 din capete = val derivatei

[Cubic Spline Interpolation] Given a function f defined on $[a, b]$ and a set of nodes, $a = x_0 < x_1 < \cdots < x_n = b$, a cubic spline interpolant, S, for f is a function that satisfies the following conditions:

(a) For each $j = 0, 1, \ldots, n-1$, $S(x)$ is a cubic polynomial, denoted by $S_j(x)$, on the subinterval $[x_j, x_{j+1})$.

(b) $S(x_j) = f(x_j)$ for each $j = 0, 1, \ldots, n$.

(c) $S_{j+1}(x_{j+1}) = S_j(x_{j+1})$ for each $j = 0, 1, \ldots, n-2$.

(d) $S'_{j+1}(x_{j+1}) = S'_j(x_{j+1})$ for each $j = 0, 1, \ldots, n-2$.

(e) $S''_{j+1}(x_{j+1}) = S''_j(x_{j+1})$ for each $j = 0, 1, \ldots, n-2$.

(f) One of the following sets of boundary conditions is satisfied:

(i) $S''(x_0) = S''(x_n) = 0$ (natural or free boundary);

(ii) $S'(x_0) = f'(x_0)$ and $S'(x_n) = f'(x_n)$ (clamped boundary).

Consider the points $(x^{[k]}, y^{[k]}) = (1, 1), (2, 5), (3, 4)$. We seek to fit a cubic polynomial on the interval [1, 2] and another cubic polynomial on the interval [2, 3]. These take the forms

$$p_1(x) = \delta_1 x^3 + \gamma_1 x^2 + \beta_1 x + \alpha_1 \quad [2.117]$$

$$p_2(x) = \delta_2 x^3 + \gamma_2 x^2 + \beta_2 x + \alpha_2 \quad [2.118]$$

Our first condition requires

$$p_1(1) = 1 \quad [2.119]$$

$$p_1(2) = 5 \quad [2.120]$$

$$p_2(2) = 5 \quad [2.121]$$

$$p_2(3) = 4 \quad [2.122]$$

The second condition requires

$$p_1'(2) = p_2'(2) \quad [2.123]$$

The third condition requires

$$p_1''(2) = p_2''(2) \quad [2.124]$$

Finally, the last condition requires

$$p_1''(1) = 0 \quad [2.125]$$

$$p_2''(3) = 0 \quad [2.126]$$

We have eight equations in eight unknowns. These can be expressed as

We have eight equations in eight unknowns. These can be expressed as

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 8 & 4 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 8 & 4 & 2 & 1 \\ 0 & 0 & 0 & 0 & 27 & 9 & 3 & 1 \\ 12 & 4 & 1 & 0 & -12 & -4 & -1 & 0 \\ 12 & 2 & 0 & 0 & -12 & -2 & 0 & 0 \\ 6 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 18 & 2 & 0 & 0 \end{pmatrix} \begin{pmatrix} \delta_1 \\ \gamma_1 \\ \beta_1 \\ \alpha_1 \\ \delta_2 \\ \gamma_2 \\ \beta_2 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 5 \\ 4 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad [2.127]$$

which we solve to obtain

$$\begin{pmatrix} \delta_1 \\ \gamma_1 \\ \beta_1 \\ \alpha_1 \\ \delta_2 \\ \gamma_2 \\ \beta_2 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} -1.25 \\ 3.75 \\ 1.50 \\ -3.00 \\ 1.25 \\ -11.25 \\ 31.50 \\ -23.00 \end{pmatrix} \quad [2.128]$$

Our two polynomials are

$$p_1(x) = -1.25x^3 + 3.75x^2 + 1.50x - 3.00 \quad [2.129]$$

$$p_2(x) = 1.25x^3 - 11.25x^2 + 31.50x - 23.00 \quad [2.130]$$

sursa:http://www.value-at-risk.net/cubic-spline-interpolation/#exercise_2_17
curbe bezier: https://www.youtube.com/watch?v=dXECQRlmlaE

Polinoame Bernstein: Teorie, pagina 10

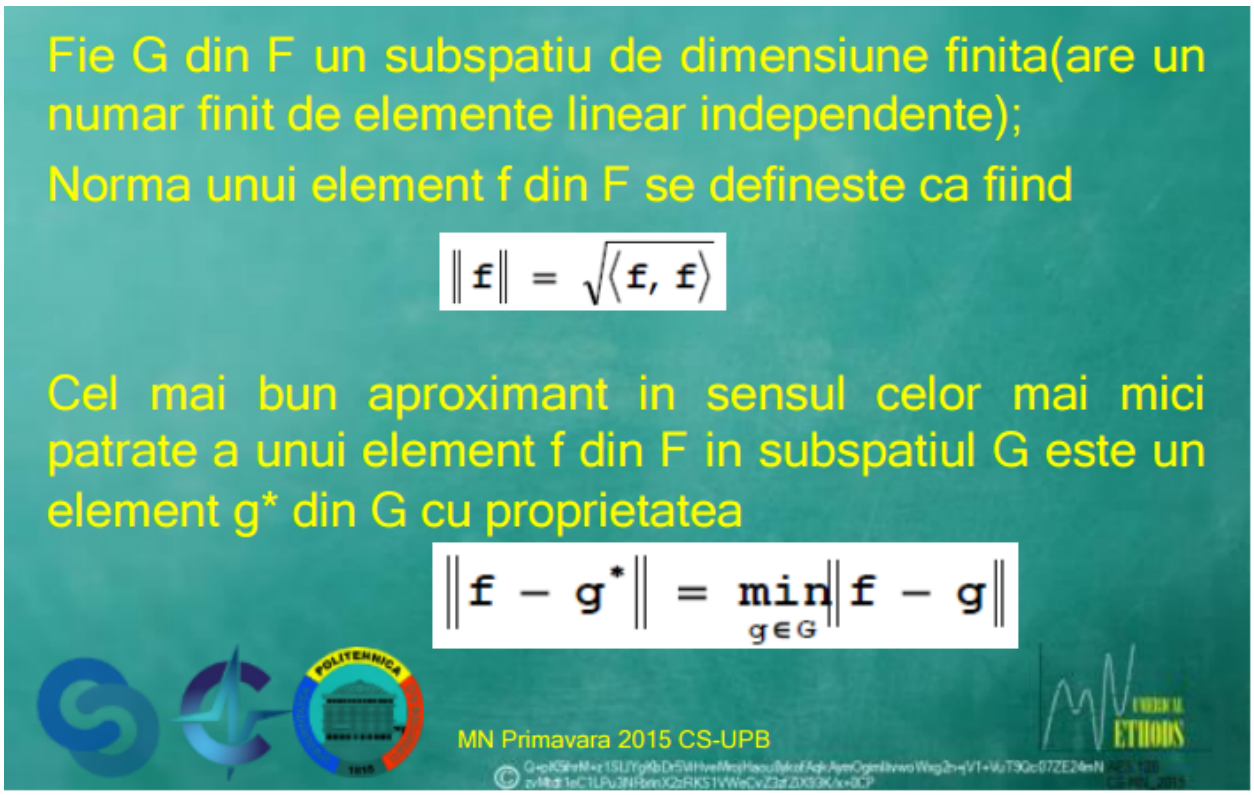
Curs 10:

Aproximari: Interpolarea presupune ca functia sa treaca prin cele n puncte x_i; totusi, datorita propagarii erorilor de masurare, acest lucru nu este fezabil. Astfel, vrem sa aproximam valorile functiilor.

$$E_2(a_0, a_1) = \sum_{i=1}^{10} |y_i - (a_1 x_i + a_0)|^2 = \sum_{i=1}^{10} (y_i - (a_1 x_i + a_0))^2 .$$

Q:Ce presupune sistemul Gram?

A:



Aproximare in sensul CMMP

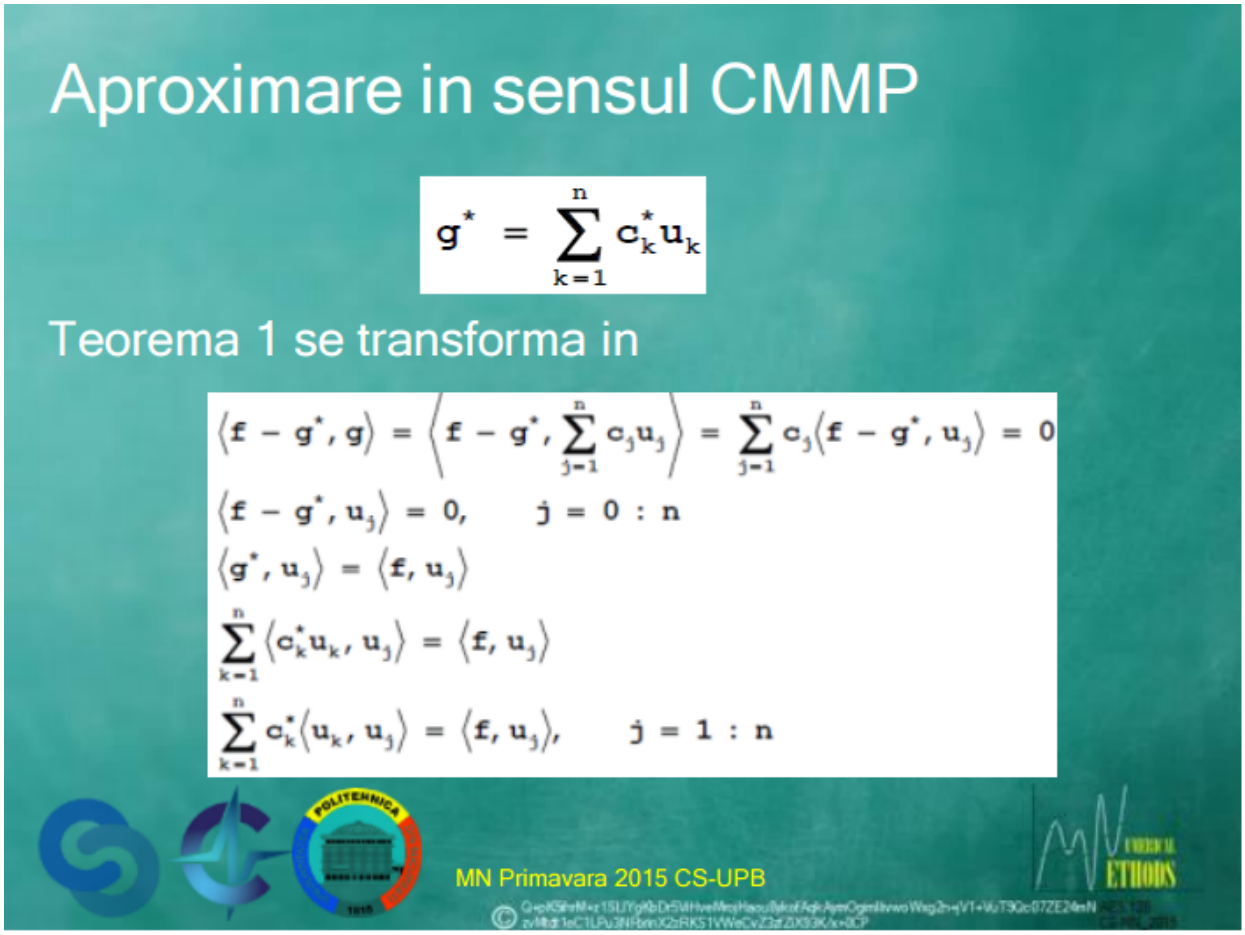
T1: g* din G este cel mai bun aproximant al lui f din F daca $\langle f-g^*, g\rangle=0, \ \forall g\in G$

Aproximare in sensul CMMP

De unde rezulta un sistem de ecuatii

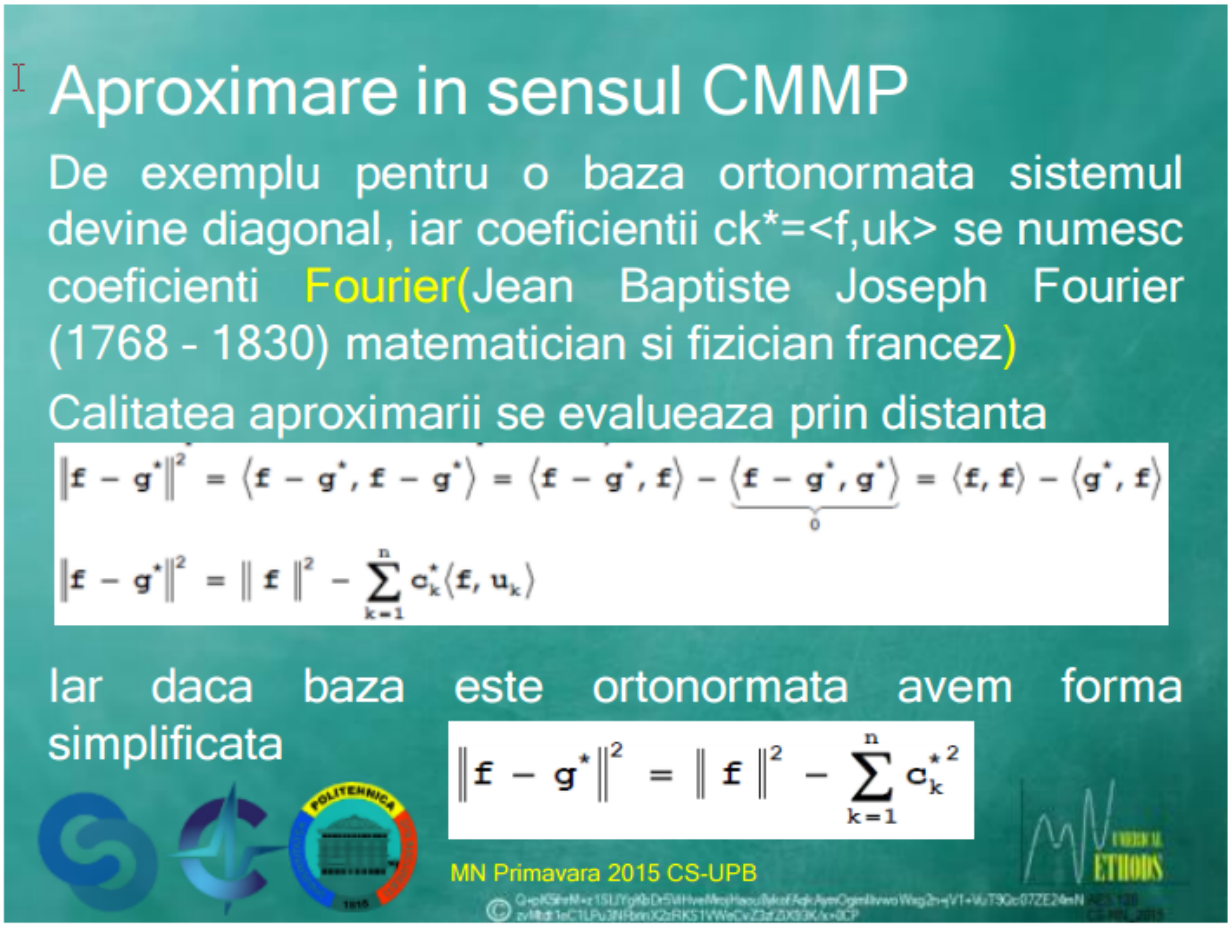
$$\begin{cases} \langle u_1, u_1\rangle \cdot c_1 + \langle u_2, u_1\rangle \cdot c_2 + \ldots + \langle u_n, u_1\rangle \cdot c_n = \langle f, u_1\rangle \\ \langle u_1, u_2\rangle \cdot c_1 + \langle u_2, u_2\rangle \cdot c_2 + \ldots + \langle u_n, u_2\rangle \cdot c_n = \langle f, u_2\rangle \\ \ldots \quad\quad\quad \ldots \quad\quad\quad \ldots \quad\quad\quad \ldots \\ \langle u_1, u_n\rangle \cdot c_1 + \langle u_2, u_n\rangle \cdot c_2 + \ldots + \langle u_n, u_n\rangle \cdot c_n = \langle f, u_n\rangle \end{cases}$$

Sistem simetric si rau conditionat, al carui determinant poarta munele de determinat GRAM. Deoarece rezolvarea sistemului este anevoioasa se aleg cazuri particulare, adica diferite baze.



Q:Care sunt coeficientii Fourier?

A:



Q:La Transformata Fourier Rapida, cum se face trecerea de la o iteratie la alta?(pentru exemplul din laborator,coeficientii c,d,e,f)

A:

Transformata Fourier Rapida FFT

La aproximare discreta trigonometrica in sensul cmmp coeficientii polinomului g* se obtineau din

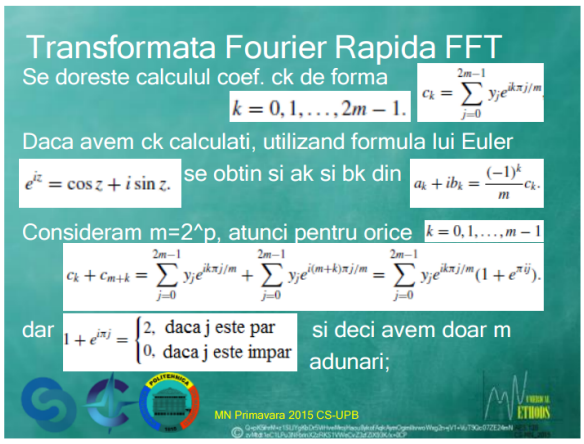
$$a_k = \frac{1}{m}\sum_{j=0}^{2m-1} y_j \cos kx_j, \quad k = 0, 1, \ldots, m,$$

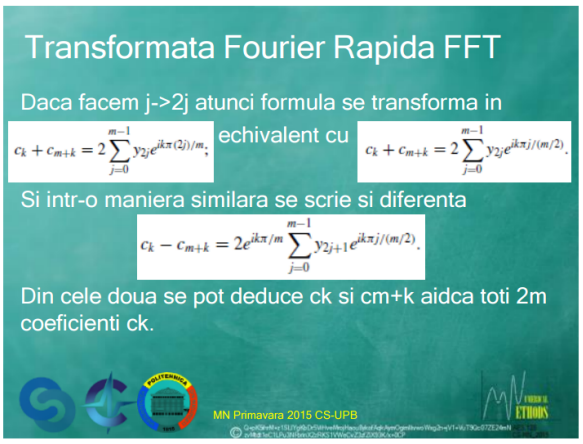
$$b_k = \frac{1}{m}\sum_{j=0}^{2m-1} y_j \sin kx_j \quad k = 1, 2, \ldots, m-1.$$

Se doreste obtinea coeficientilor mai eficient(aici avem 4m^2 operatii) prin calculul altor coeficienti.



MN Primavara 2015 CS-UPB





Q: pentru ce se calculeaza d,e,f?

Curs 11: Derivare si integrare numerica:

Legatura Lagrange-derivare: Formula pentru derivare se obtine derivand expresia polinomului Lagrange cu tot cu eroare.; integrarea se integrand polinomul Lagrange.
http://cs.curs.pub.ro/2014/pluginfile.php/14028/mod_resource/content/0/lab_mn_11.pdf
(foarte bine explicat in laborator)

[Linear Least Squares]
The linear least squares solution for a given collection of data $\{(x_i, y_i)\}_{i=1}^m$ has the form $y = a_1 x + a_0$, where

$$a_0 = \frac{\left(\sum_{i=1}^m x_i^2\right)\left(\sum_{i=1}^m y_i\right) - \left(\sum_{i=1}^m x_i y_i\right)\left(\sum_{i=1}^m x_i\right)}{m\left(\sum_{i=1}^m x_i^2\right) - \left(\sum_{i=1}^m x_i\right)^2}$$

and

$$a_1 = \frac{m\left(\sum_{i=1}^m x_i y_i\right) - \left(\sum_{i=1}^m x_i\right)\left(\sum_{i=1}^m y_i\right)}{m\left(\sum_{i=1}^m x_i^2\right) - \left(\sum_{i=1}^m x_i\right)^2}.$$

Curs 12:
Extrapolare Richardson:
Aproximarea unei valori M cu o functie N1(h) ce depinde de un parametru h, de obicei pasul.Metoda functioneaza prin injumatatirea pasului de aproximare h.

Integrare Romberg:

Quadraturi adaptive:

over a large interval into integrals over smaller subintervals. The approach used in that section involved equally-sized subintervals, which permitted us to combine the individual approximations into a convenient form. Although this is satisfactory for most problems, it leads to increased computation when the function being integrated varies widely on some, but not all, parts of the interval of integration. In this case, techniques that adapt to the differing accuracy needs of the interval are superior. Adaptive numerical methods are particularly popular for inclusion in professional software packages since, in addition to being efficient, they generally provide approximations that are within a given specified tolerance.

for some number ξ in (a, b), where we denote the Simpson's rule approximation on $[a, b]$ by

$$S(a, b) = \frac{h}{3}[f(a) + 4f(a + h) + f(b)].$$

[Adaptive Quadrature Error Estimate] If

$$\left| S(a, b) - S\left(a, \frac{a+b}{2}\right) - S\left(\frac{a+b}{2}, b\right) \right| \leq 15\varepsilon,$$

then

$$\left| \int_a^b f(x)\, dx - S\left(a, \frac{a+b}{2}\right) - S\left(\frac{a+b}{2}, b\right) \right| \lessapprox \varepsilon.$$